

<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC FEB MAR

23

2017 2018 2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018

~~Particle~~

Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



How to remove Particle from your Digispark Oak ESP8266

9 months ago

esp8266 · digispark · oak · particle

So a while back I backed the [Digistump Oak Kickstarter](#). I thought these boards were great and I'd supported other Digistump projects by backing or purchasing Erik's dev boards.

One of the big drawcards was over the air updates via

Subscribe via RSS

Subscribe via email



© Making a Mess 2018

Ghost & StayPuft

<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC FEB MAR

23

2017 2018 2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



Subscribe via RSS

Subscribe via email



© Making a Mess 2018

Ghost & StayPuft

Arduino Core didn't have great OTA support so this was win-win. (It's now waaaaay better.)

Now the boards are great and the Oak support for arduino and the firmware was solid. The one thing I never found consistantly solid however was the Particle OTA support. I'd be able to flash it successfully a few times, come back a few months later and then flashing would just randomly and annoyingly fail.

Each time it failed I'd spend days troubleshooting, manually wiping the firmware/bootloader and replacing it until it kinda worked. The Oaks would phone home properly to Particle and show up on the Dashboard but would either stop

<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC FEB MAR

23

2017 2018 2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



responding.

Eventually when I came back to work on a project I'd built on the Oak pinout I finally gave in as I spent all the time I'd intended to use for development on fault finding the Particle OTA.

At its core, the Oak is simply an ESP8266 12E with a 4mb flash on it. I've had plenty of experience flashing them using esptool.py and esptool-ck, so I did what any self respecting person did.

Firstly, I wired a Serial -> USB Adaptor to it:

<https://digistump.com/wiki/oak/tutorials/serialfirmware> was a good primer for how

Subscribe via RSS

Subscribe via email



© Making a Mess 2018

Ghost & StayPuft

<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC FEB MAR

23

2017 2018 2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



Make sure you connect P2 to ground, or the ESP8266 will not boot into serial flash mode.



My setup looks like this.

Then I backed it up using esptool.py:

```
python esptool.py --  
baud 115200 --port  
/dev/tty.SLAB_USBtoUA  
RT read_flash  
0x000000 0x400000  
flash_devicenumbe.bi  
n
```

You will definitely need to adjust the --port part of this command to suit your circumstances (COM on windows, usually /dev/tty. on mac and similar on linux) and make sure you put the device number or a friendly

Subscribe via RSS

Subscribe via email



© Making a Mess 2018

Ghost & StayPuft

<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC FEB MAR

23

2017 2018 2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



have to go back
you're not totally
screwed.

Then, I wiped
it with
esptool.py:

```
python esptool.py --  
baud 115200 --port  
/dev/tty.SLAB_USBtoUA  
RT erase_flash
```

Again, change the
port to whatever port
your serial-usb
adaptor shows up as
on your relevant OS.

Your
Digispark Oak
is now
Particle free

But it's still not
entirely usable. At
this point you could:

- Flash it using the
[Arduino](#)
[ESP8266 Core](#)
and use
[ESP8266 OTA](#)

Subscribe via RSS

Subscribe via email



© Making a Mess 2018

Ghost & StayPuft

<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC FEB MAR

23

2017 2018 2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



Subscribe via RSS

Subscribe via email



© Making a Mess 2018

Ghost & StayPuft

[Micropython](#),
[Espruino](#) or half
a dozen other
projects.

I expect most people
will want the first
option as it means
you can use local or
HTTP based Over The
Air updating.

But we need to tell
the Arduino ESP8266
Core about the
Digispark Oak as the
pin numbers on the
board don't match
what pins are being
used on the ESP8266.

Telling Arduino about Oak's pins.

First thing you'll need
to do is to create a
variant in the
hardware folder for
the ESP8266 core.
These instructions
are assuming you've
already [installed the
core](#).

<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC FEB MAR

23

2017 2018 2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



Subscribe via RSS

Subscribe via email



© Making a Mess 2018

Ghost & StayPuft

Documents folder.

The full address will look something like:

```
C:\Users\Username\Documents\ArduinoData\packages\esp8266\hardware\esp8266\2.3.0\variants
```

On Mac, it's in your user's Library folder, so the location is:

```
/Users/<USERNAME>/Library/Arduino15/packages/esp8266/hardware/esp8266/2.3.0/variants
```

The version numbers may change when new releases of the core come out, so you will need to double-check.

To begin, [unzip this zip file in the variants folder](#). This should create a folder called DigiStumpOak with a file inside it called Pins_Arduino.h. For those of you who'd rather not download



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search

Q

 Subscribe via RSS

 Subscribe via email

    

© Making a Mess 2018

 Ghost & StayPuft

Then, we will need to add some lines to boards.txt, which is in the folder above the variants folder.

At the end of the boards.txt file, add the following:

1	#####
2	oak.name=Digistump C
3	
4	oak.upload.tool=espt
5	oak.upload.speed=115
6	oak.upload.resetmeth
7	oak.upload.maximum_s
8	oak.upload.maximum_c
9	oak.upload.wait_for_
10	oak.serial.disableDT
11	oak.serial.disableRT
12	
13	oak.build.mcu=esp826
14	oak.build.f_cpu=8000
15	oak.build.board=ESP8
16	oak.build.core=esp82
17	oak.build.variant=Di
18	oak.build.flash_mode
19	oak.build.spiffs_pag
20	oak.build.debug_port
21	oak.build.debug_leve
22	
23	oak.menu.CpuFrequenc
24	oak.menu.CpuFrequenc
25	oak.menu.CpuFrequenc
26	oak.menu.CpuFrequenc
27	
28	oak.menu.FlashFreq.4
29	oak.menu.FlashFreq.4



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search

Q

📡

Subscribe via RSS

✉

Subscribe via email

🔄

in

🐦

📺

✉

© Making a Mess 2018

☰ Ghost & StayPuft

33	oak.menu.FlashMode.c
34	oak.menu.FlashMode.c
35	oak.menu.FlashMode.c
36	oak.menu.FlashMode.c
37	oak.menu.FlashMode.c
38	oak.menu.FlashMode.c
39	oak.menu.FlashMode.c
40	oak.menu.FlashMode.c
41	
42	oak.menu.UploadSpeed
43	oak.menu.UploadSpeed
44	oak.menu.UploadSpeed
45	oak.menu.UploadSpeed
46	oak.menu.UploadSpeed
47	oak.menu.UploadSpeed
48	oak.menu.UploadSpeed
49	oak.menu.UploadSpeed
50	oak.menu.UploadSpeed
51	oak.menu.UploadSpeed
52	oak.menu.UploadSpeed
53	oak.menu.UploadSpeed
54	oak.menu.UploadSpeed
55	oak.menu.UploadSpeed
56	oak.menu.UploadSpeed
57	oak.menu.UploadSpeed
58	oak.menu.UploadSpeed
59	oak.menu.UploadSpeed
60	
61	oak.menu.FlashSize.4
62	oak.menu.FlashSize.4
63	oak.menu.FlashSize.4
64	oak.menu.FlashSize.4
65	oak.menu.FlashSize.4
66	oak.menu.FlashSize.4
67	oak.menu.FlashSize.4
68	oak.menu.FlashSize.4
69	
70	oak.menu.FlashSize.4
71	oak.menu.FlashSize.4
72	oak.menu.FlashSize.4
73	oak.menu.FlashSize.4
74	oak.menu.FlashSize.4



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search

Q

Subscribe via RSS

Subscribe via email

© Making a Mess 2018

Ghost & StayPuft

78	oak.menu.ResetMethod
79	oak.menu.ResetMethod
80	oak.menu.ResetMethod
81	oak.menu.ResetMethod
82	
83	oak.menu.Debug.Disab
84	oak.menu.Debug.Disab
85	oak.menu.Debug.Seria
86	oak.menu.Debug.Seria
87	oak.menu.Debug.Seria
88	oak.menu.Debug.Seria
89	
90	oak.menu.DebugLevel.
91	oak.menu.DebugLevel.
92	oak.menu.DebugLevel.
93	oak.menu.DebugLevel.
94	oak.menu.DebugLevel.
95	oak.menu.DebugLevel.
96	oak.menu.DebugLevel.
97	oak.menu.DebugLevel.
98	oak.menu.DebugLevel.
99	oak.menu.DebugLevel.
100	oak.menu.DebugLevel.
101	oak.menu.DebugLevel.
102	oak.menu.DebugLevel.
103	oak.menu.DebugLevel.
104	oak.menu.DebugLevel.
105	oak.menu.DebugLevel.
106	oak.menu.DebugLevel.
107	oak.menu.DebugLevel.
108	oak.menu.DebugLevel.
109	oak.menu.DebugLevel.
110	oak.menu.DebugLevel.
111	oak.menu.DebugLevel.
112	oak.menu.DebugLevel.
113	oak.menu.DebugLevel.
114	oak.menu.DebugLevel.
115	oak.menu.DebugLevel.
116	oak.menu.DebugLevel.
117	oak.menu.DebugLevel.
118	oak.menu.DebugLevel.
119	oak.menu.DebugLevel.

<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC FEB MAR

23

2017 2018 2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018

[addtoboards.txt](#) [view raw](#)
hosted with by GitHub

Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



Congrats, you're done!

At this point, you can now reopen the Arduino IDE and you should end up seeing another board in your Boards menu under Tools.

I may submit this upstream as a new board definition for the Arduino ESP8266 core, but that will take a little bit to be added.

If you've found this useful, please let me know.

Subscribe via RSS

Subscribe via email



© Making a Mess 2018

Ghost & StayPuft

B
r
e
n
d
a
n
'
n
o
g

Share this post



<http://nog3.net/2017/06/05/removing-particle-from-oak>

Go

DEC

FEB

MAR



23



2017

2018

2019



▼ About this capture

[10 captures](#)

17 Nov 2017 - 23 Feb 2018



Making a Mess

This is an attempt at project documentation by Brendan 'nog3' Halliday.

Search



📍 *Brisbane,
Australia*



Subscribe via RSS



Subscribe via email



© Making a Mess 2018

☰ Ghost & StayPuft

comments powered
by Disqus